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Abstract

Significance

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Keyword 1 | Keyword 2 | Keyword 3 | ...

In the classical Darwinian Paradigm, individuals whose phenotype is favored by selection has a better chance of survival and of contributing their genes to the offspring population (1). Many quantitative genetic models built on this paradigm assume a direct genotype-phenotype mapping, in which an individual's phenotype is determined by its own genotype and a generic, non-social environment (2). However, this assumption is challenged once social interactions are taken into account, because part of the environment is made up of the genotypes and phenotypes of interacting conspecifics (3, 4). When organisms interact with each other, certain phenotype may be affected depending on who the interacting partner is and what the nature of the interaction is. For example, in a cooperative setting, reciprocity in cooperative breeding shows that a bird that was helped is more likely to help the helper in the future, potentially enhancing both individual's reproductive outcome (5); in a conflict setting, a subordinate fish displays more aggressive and courtship behavior when the audience consists of other subordinantes, but suppresses both behaviors when a dominant male is watching (6, 7). Such a phenotype is termed a social interacting phenotype, a trait whose expression is affected by interactions with conspecifics (3).

In order to understand the evolutionary consequences of such social interacting phenotype, Moore et al. proposed a quantitative genetic model that partitioned the phenotype into a direct genetic contribution from one's own genotype, and an indirect genetic contribution from a social partner genotype (3). They predicted that the evolutionary response of a social

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interacting phenotype should differ, often be greater, compared to a nonsocial phenotype, especially when the same trait is affected between social partners reciprocally, creating a feedback loop that drives phenotype evolution. Many extensions of this model have since been developed within the quantitative genetic framework (8–11), and studies have estimated the value of the proposed interaction coefficient, ψ , in empirical systems (12–14).

However, the field of population genetics argues that sociality in general should relax selection instead of intensifying it. The theory assumes that social structure segments a population into breeding groups characterized by reproductive skew, in which few individuals contribute to the offspring population (15). Group formation leads to both higher inbreeding than a well-mixed population, and fewer breeding individuals. Both structural changes lead to stronger drift and less effective selection at genome-wide level (15, 16). It has been shown in spiders and ants that there is genome-wide elevated molecular evolution in social species compared to nonsocial relatives, suggesting relaxed selection (17–22).

Such debate between relaxed selection or accelerated adaptation also exist in empirical work. As a general example, immune genes are expected to be under different selection for social living species relative to solitary ones. Social interactions, especially group living, increases contact rate and relatedness among individuals, providing the ideal condition for disease transmission (23, 24). Immune health depends on both genotype and environmental condition such as pathogen exposure in early life, and social interactions such as grooming and hygienic behaviors (25, 26). In ants and honeybees, it has been shown that immune genes have elevated dN/dS relative to *Drosophila* immune genes or non-immune genes in bees, which can be a signature of either relaxed or positive selection (27). Studies have shown support for both sides: enrichment of immunity genes for positive selection was shown in (28), while (29) provided support for relaxed purifying selection. Within the same broad functional category and same selective pressure cause from social living, there is no convergent agreement on whether the dominant signature is relaxed constraint or adaptive acceleration, and the conclusion is gene-, taxon-, and method-dependent (30, 31).

While immune genes show both positive and relaxed selection in parallel, another gene family shows a different selective pattern when it is affected by the highest level of sociality. Eusocial species evolved division of labor where members of the colony is divided into different castes with specific tasks such as foraging or nursing. Besides genetics, an individual's caste in adulthood also depends on the social interactions: diet received in larval stage, chemical signals released by other members, and age or immediate need of the colony (32). These are important factors for an individual to develop or switch to a specific caste. Given different social groups, genes that are caste-biased are expressed differentially and has shown elevated rate of evolution compared to noncaste-biased genes, similar to that of immune genes (33). Unlike immune genes, the evidence suggested that the caste-biased genes may undergo relaxed constraint followed by caste-specific positive selection (28, 33). Most of the caste-associated genes are lineage-dependent and evolutionarily younger, suggesting caste-specific positive selection following the

evolution of eusociality and division of labor (34). Rather than both modes of selection happening in parallel, these socially influenced genes in eusocial species suggest that both modes can happen sequentially, adding complexity beyond what previous theory predicted.

Across theoretical and empirical studies, the same pattern emerges where a socially influenced trait evolves faster or slower is not predicted by whether a species is social, but by the specific structure of the interaction. Another field that similarly stresses the structure of social interactions is social learning theory. Rogers' paradox is a foundational result in this literature, stating that unconditional social learning does not raise mean population fitness; instead, the fitness payoff depends on who, when, and what individuals copy, formalized as social learning strategies (35–37). Social learning strategies further decompose social group by specifying who is copied, offering a more fine-grained perspective than the coarser sociality measures used in the theories discussed above. In this study, we incorporate social learning strategies and explicit genetic architecture into a social interaction model extended from Moore et al. (3). Given the complex, structure-dependent selection patterns observed in empirical systems, we test social populations under different selection modes in an attempt to characterize how different social learning strategies shape population-level evolutionary outcomes.

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